PROJECT AND TEAM INFORMATION

Project Title

Student Skill Roadmap

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

In today's academic environment, B.Tech CSE students often struggle to manage their academic responsibilities and career development simultaneously. Skill learning, coding practice, assignments, and study materials are usually scattered across multiple platforms, making it difficult to maintain consistency and track progress effectively. Many students lack a clear roadmap that guides them on what to learn and in what sequence for career growth. Additionally, deadlines and coding practice tracking are often managed manually, leading to inefficiency and missed opportunities.
The motivation behind this project is to create a centralized web-based platform that integrates skill development tracking, assignment management, and coding practice access into a single system. By providing structured roadmaps, organized academic management, and streamlined platform linking, this project aims to enhance productivity, improve clarity in career planning, and promote disciplined learning among students.

State of the Art / Current solution

Currently, students rely on multiple independent platforms to manage their academic and career development activities. Learning roadmaps are available on websites like blogs, YouTube channels, and online communities, but they are not personalized or progress-tracked within a single system. Assignment management is typically handled through tools such as Google Classroom, email, or manual notes, which focus only on academic tasks and not overall skill growth. Coding practice is done on platforms like LeetCode, Codeforces, and HackerRank, but these platforms operate independently and do not integrate with academic management tools.

There is no unified platform that combines structured skill roadmaps, assignment tracking, and centralized access to coding practice profiles in one place specifically for B.Tech CSE students. This fragmentation reduces productivity and makes it difficult for students to monitor both academic performance and career preparation efficiently within a single system.

Project Goals and Milestones

The main objective of this project is to develop a centralized student management and career development system that assists B.Tech CSE students in organizing their academic tasks and tracking their skill growth in a structured manner. The system will integrate a skill roadmap with progress tracking, an assignment management system with deadline monitoring, and a coding practice hub for linking competitive programming platforms.

Initial Milestones:

1. *Research and analysis of student academic and career management challenges*
2. *Designing the system architecture and defining module structure*

3. *Creating UI wireframes and frontend templates*
4. *Designing and implementing the relational database schema*

Intermediate Milestones:

1. *Development of authentication and user management system*
2. *Implementation of Roadmap module with progress tracking functionality*
3. *Development of Assignment management module with sorting based on deadlines*
4. *Implementation of Coding Hub module with account linking functionality*

Final Milestones:

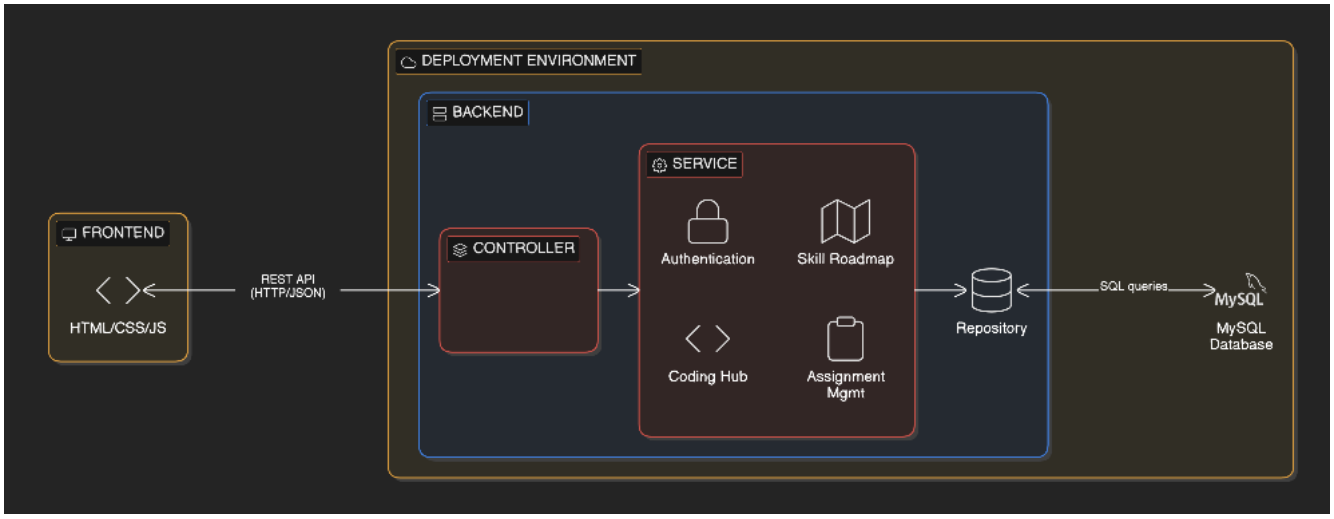
1. *Integration of frontend with backend using REST APIs*
2. *Optimization of assignment sorting and searching using DAA concepts*
3. *Testing, debugging, and performance evaluation*
4. *Final deployment and demonstration*

The project will demonstrate structured system design, modular development, and effective application of algorithmic concepts in a real-world academic management system.

Project Approach

The project will follow a structured and modular development approach based on a client-server architecture. The system will be developed using Java (Spring Boot) for the backend, MySQL as the relational database, and HTML, CSS, and JavaScript for the frontend interface.
Initially, requirement analysis will be performed to identify the core functionalities required for B.Tech CSE students. Based on this analysis, the system will be divided into independent modules such as Authentication, Skill Roadmap, Assignment Management, Coding Hub, and Dashboard.
The backend will follow a layered architecture consisting of Controller, Service, and Repository layers to ensure separation of concerns and maintainability. REST APIs will be developed to enable communication between frontend and backend using JSON data format.
The database will be designed using relational modeling principles with proper primary and foreign key relationships.
Finally, Design and Analysis of Algorithms concepts such as sorting and searching will be implemented within the Assignment module to improve efficiency and demonstrate algorithmic application.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

The final outcome of this project will be a fully functional web-based Student Management and Career Development System designed specifically for B.Tech CSE students. The system will provide a centralized platform that integrates skill tracking, assignment management, and coding practice access into a single structured application.
The key deliverables of the project include:

1. *A responsive frontend interface developed using HTML, CSS, and JavaScript.*
2. *A backend system developed using Java (Spring Boot) implementing REST APIs and layered architecture.*
3. *A relational MySQL database with properly designed tables and relationships.*
4. *Functional modules including Authentication, Skill Roadmap, Assignment Management, Coding Hub, and Dashboard.*
5. *Implementation of DAA concepts such as sorting and searching within the assignment module.*
6. *Complete project documentation including system architecture diagrams, database schema, and workflow explanation.*

The final system will demonstrate modular design, proper integration of frontend and backend, structured database management, and practical application of algorithmic concepts in a real-world academic scenario.

Assumptions

It is assumed that users have access to a stable internet connection and basic familiarity with web application usage through a computer or laptop. It is assumed that users will manually enter accurate information such as assignment details, skill progress, and coding platform usernames.
It is assumed that the system will initially handle a limited number of users and records, as it is developed for academic purposes. Therefore, scalability and high-level security mechanisms will remain basic in this phase of development.
It is also assumed that third-party coding platform data will not be integrated through official APIs in the initial version, and profile linking will be handled through username-based URL generation.
The database size is expected to remain moderate, and performance requirements will be manageable within a local development environment.

References

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